

Fraudulent Claim Detection Case Study

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PROBLEM STATEMENT

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company's current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimize financial losses and optimize the overall claims handling process.

SOLUTION METHODOLOGY

1

Importing Libraries and Data a) Import Libraries, Suppress Warnings and Set Display b) Reading the Data

2

Data Understanding and Inspection

3

Data Cleaning & Preparation a) Treatment for null values b) Handling Missing Values ,Outliers and unwanted data

4

Train Validation Split

5

EDA on training Data
a) Perform Univariate Analysis
b) Perform Bivariate Analysis

SOLUTION METHODOLOGY

6

Feature Engineering

7

Feature Scaling a) Resampling b) Creation c) handle Redundant Column
d)Combine Values in Categorical Column e)Dummy Variable Creation f) Feature Scaling

8

Model Building a) Logistic Regression b)
Random Forest Model

9

Model Evaluation

10

Prediction and Model Evaluation

DATA MANIPULATION

Total Number of Rows =1000, Total Number of Columns =40

Handled null Values and rows containing null values by dropping rows with na .

Checked value counts in each categorical column to decide whether to impute or drop missing values.

Removing columns with one unique value, irrelevant columns, and skewed categorical columns.

Identify and remove columns where a large proportion of the values are unique or near-unique, as these columns are likely to be identifiers or have very limited predictive power

13 columns data types have been updated that contain date or time information but are not stored as the appropriate data type. After Data Manipulation, we are left with 474 rows and 34 columns

Train-Validation Split

- 1.Imported libraries
- 2 Defined feature and target Variable
- 3.Split the Data

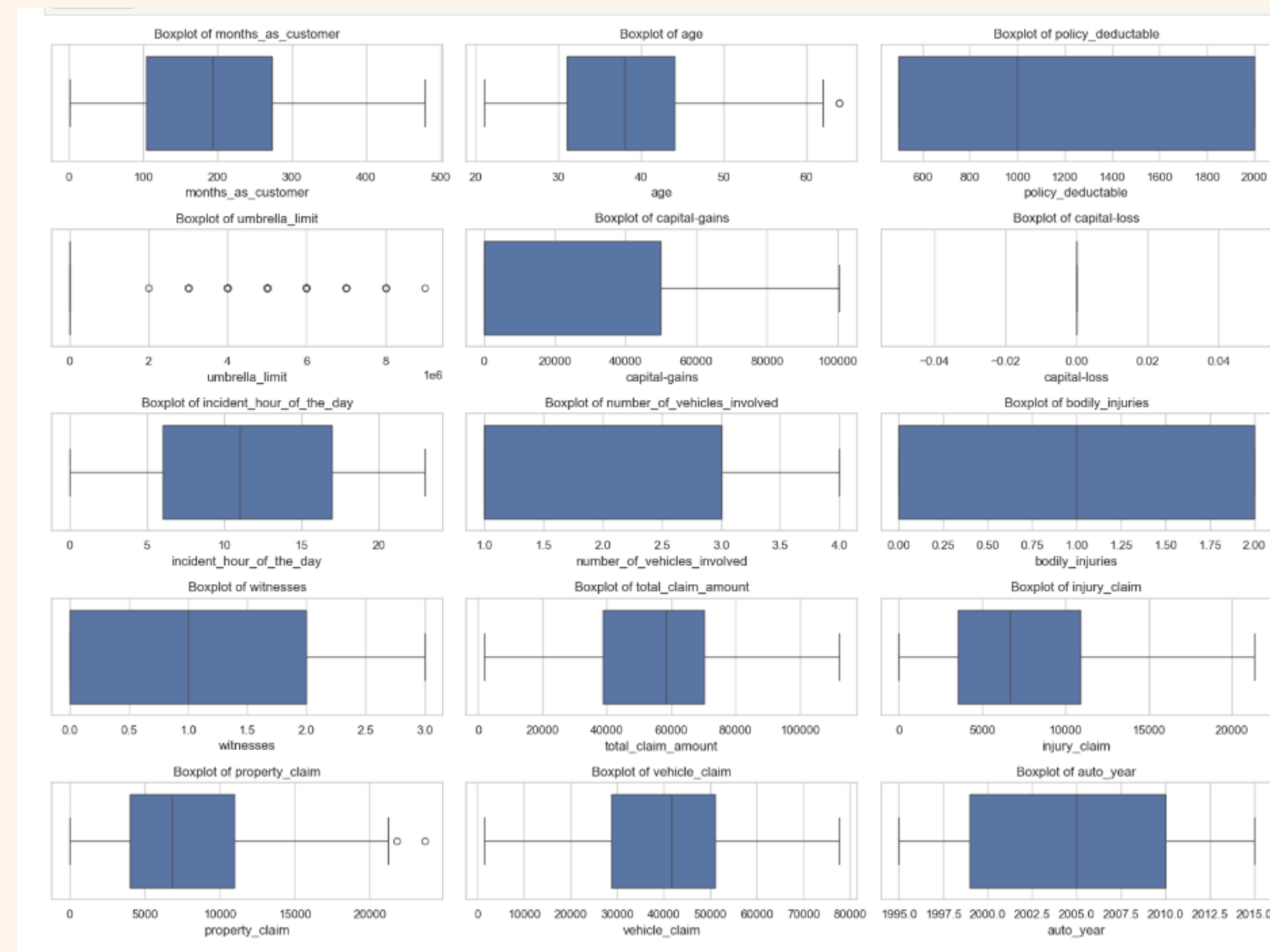
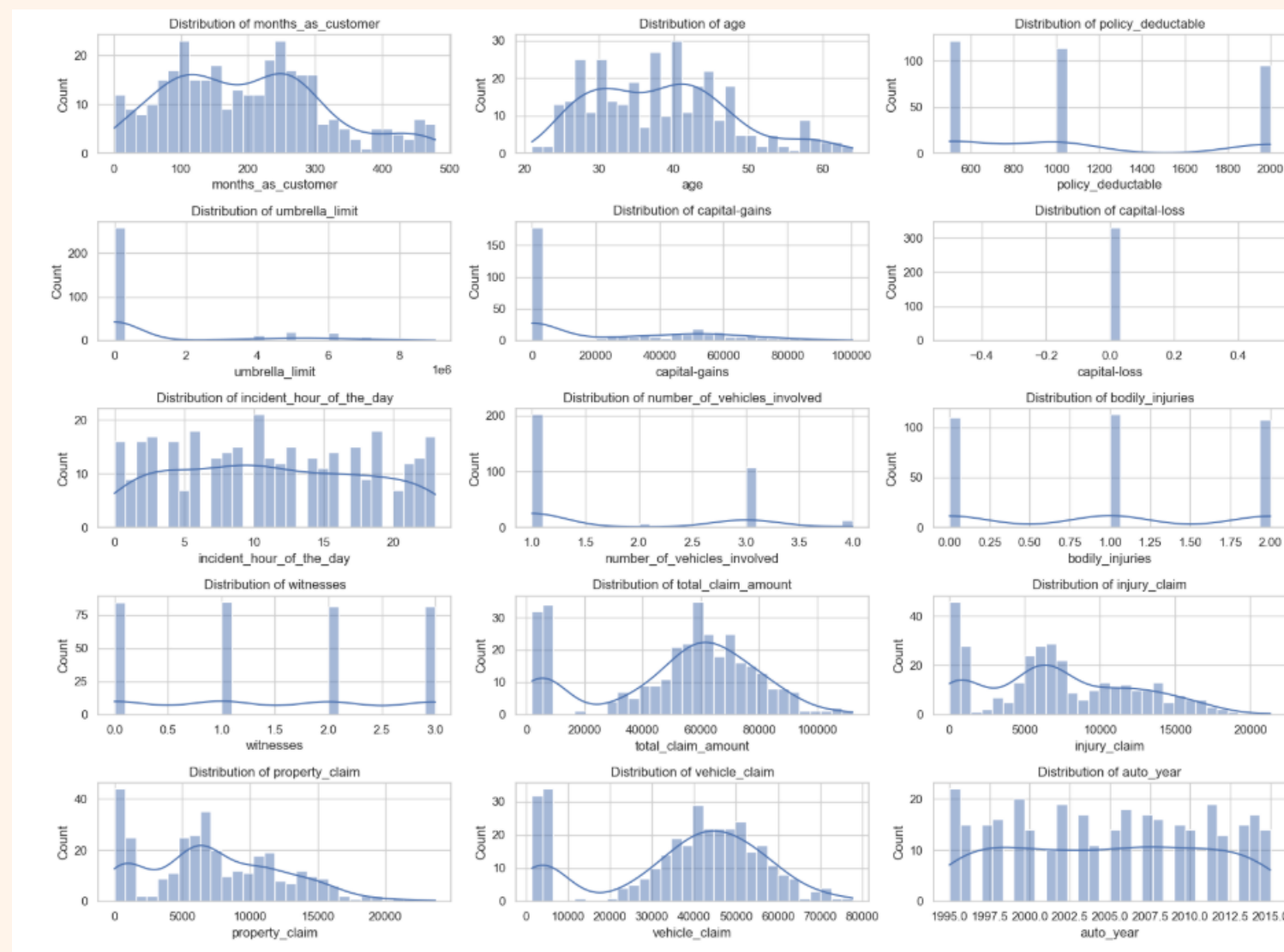
Final Test and Training data:

```
Train set shape: (331, 33)
Validation set shape: (143, 33)
Class distribution in y_train:
  fraud_reported
0      0.76435
1      0.23565
Name: proportion, dtype: float64
Class distribution in y_val:
  fraud_reported
0      0.762238
1      0.237762
Name: proportion, dtype: float64
```

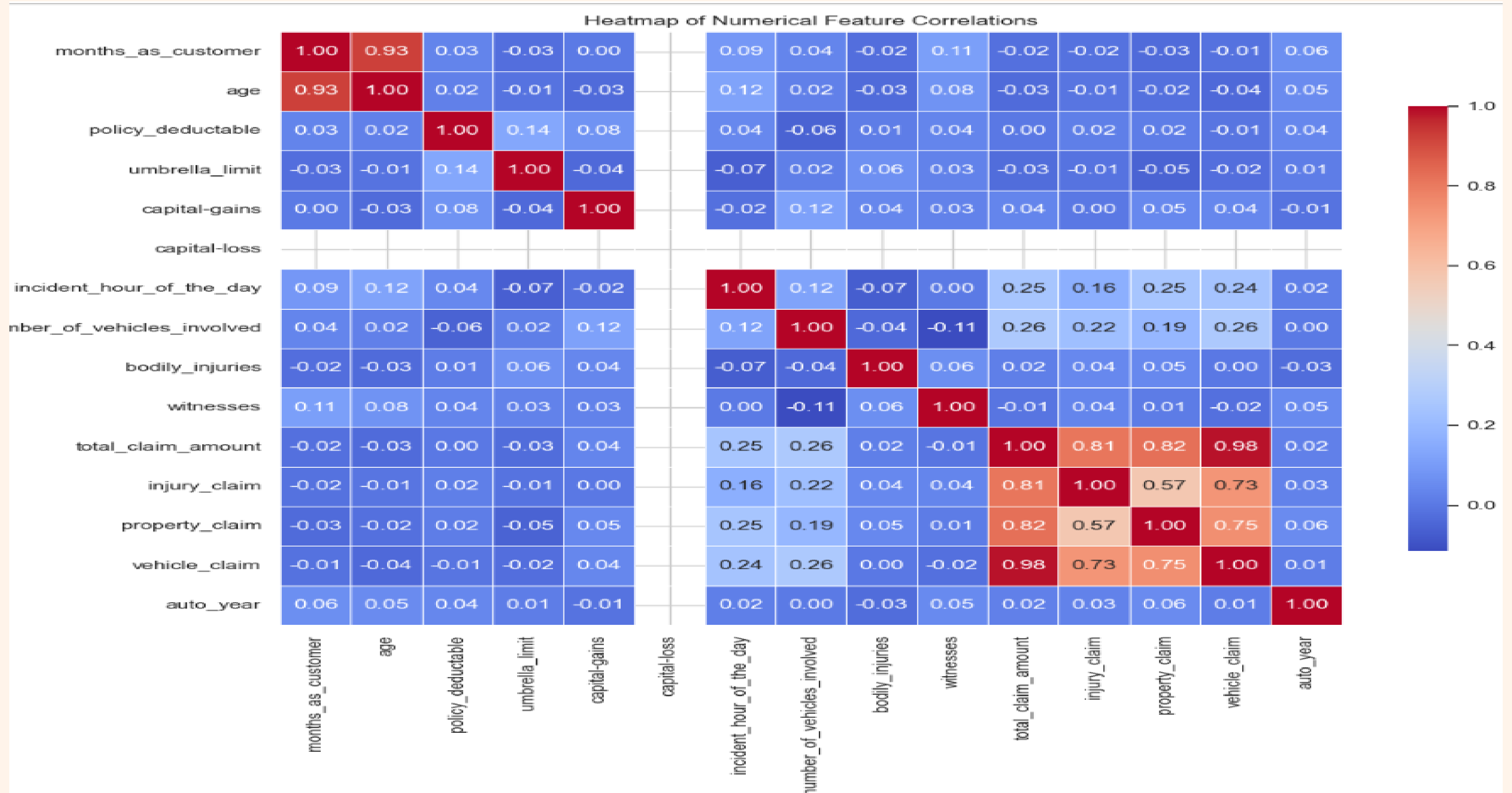
EDA

UNIVARIATE ANALYSIS

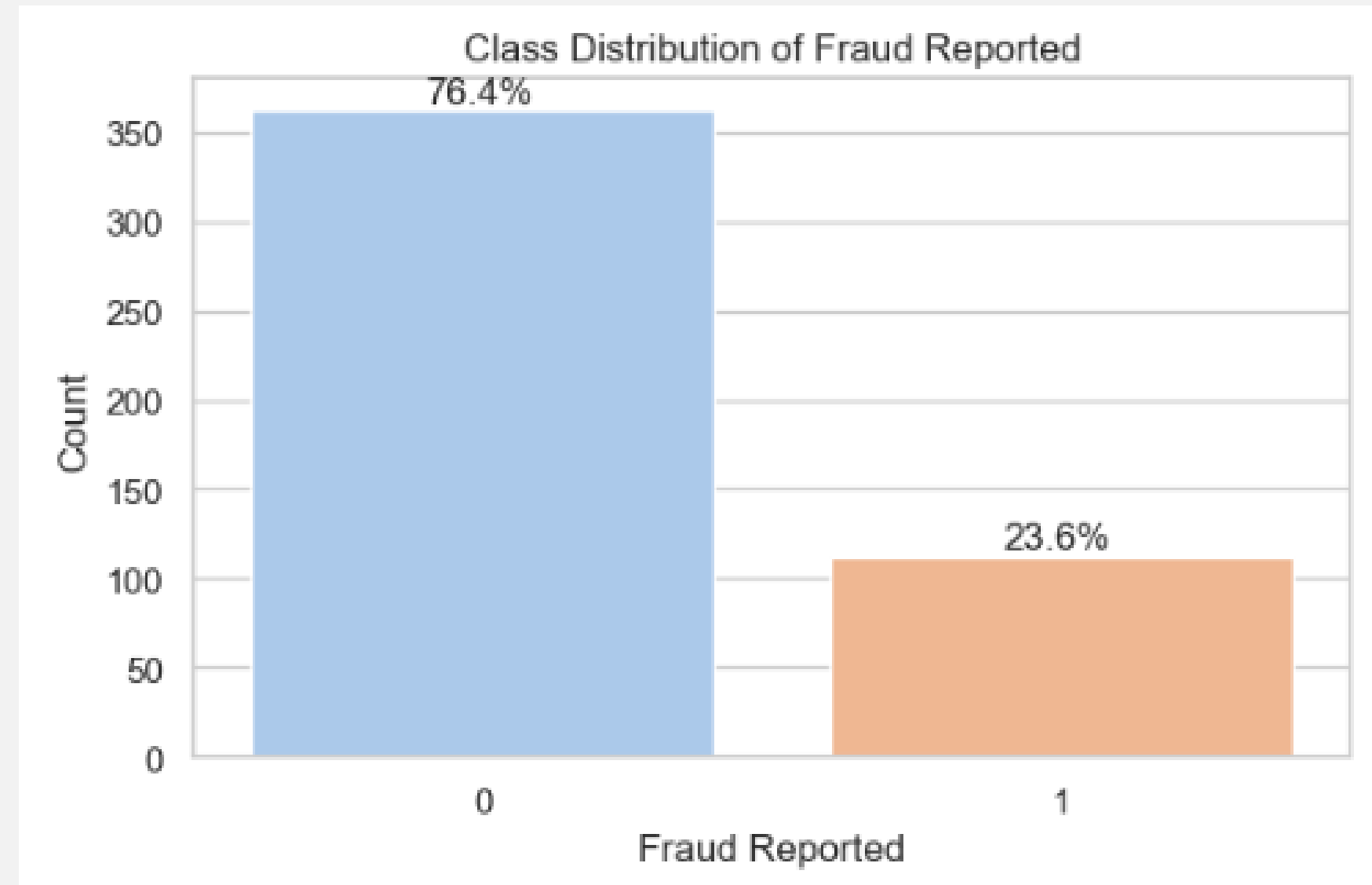
Distribution of selected Numerical Features



EDA



EDA

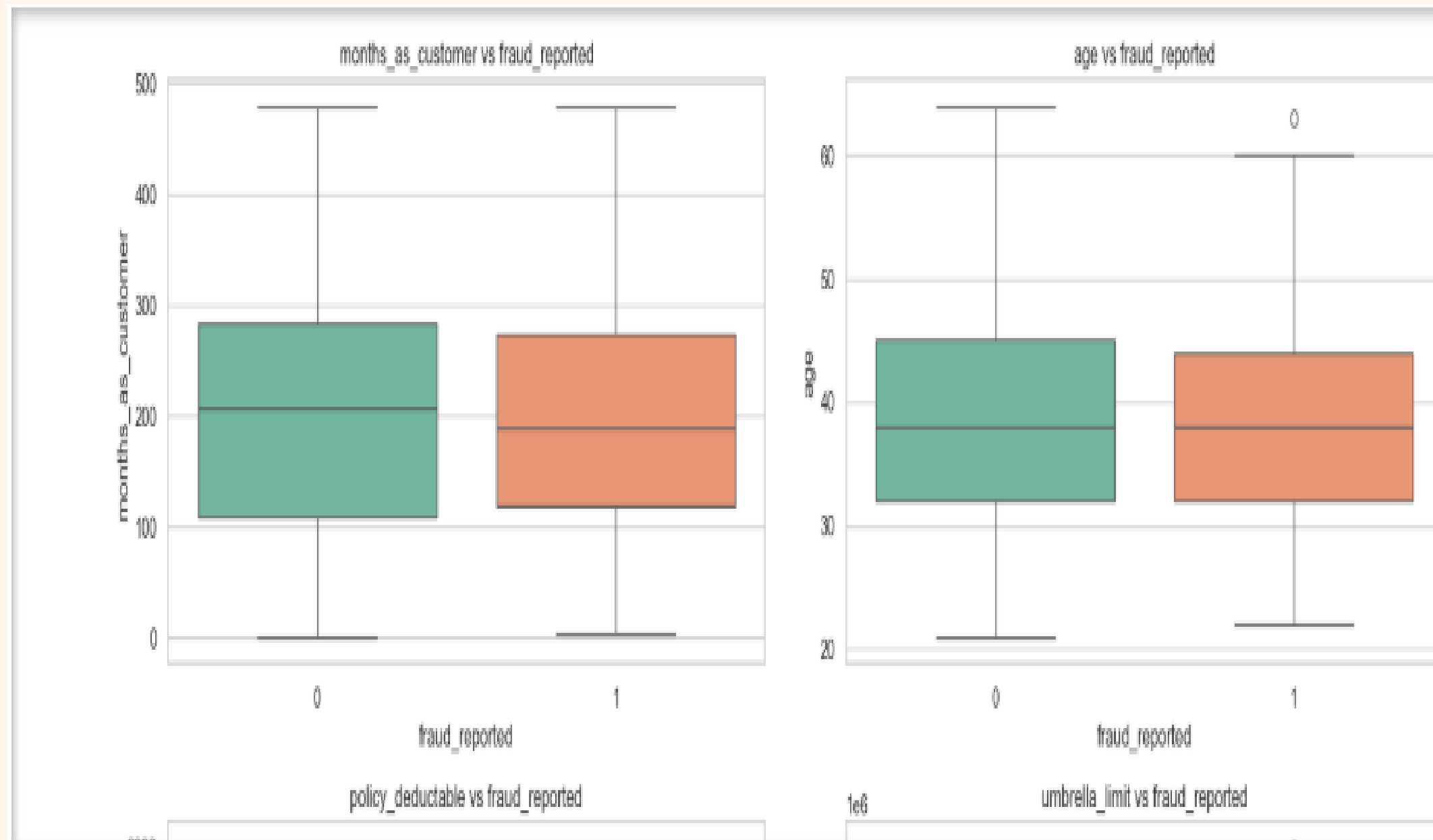


CLASS DISTRIBUTION OF FRAUD REPORTED

This univariate analysis reveals instances of reported fraud are considerably less frequent than instances where fraud was not reported.

EDA

BIVARIATE ANALYSIS



Analysis of "months_as_customer" vs "fraud_reported" states that customers who reported fraud tend to have been customers for a slightly shorter duration compared to those who did not report fraud.

Analysis of "age" vs "fraud_reported" states that there doesn't appear to be a substantial difference in the age distribution between individuals who reported fraud and those who did not. Age might not be a strong distinguishing factor for fraud reporting in this dataset.

EDA Bivariate Analysis

These initial bivariate analyses provide some interesting insights:

- The duration of being a customer ("months_as_customer") might have a subtle inverse relationship with fraud reporting – newer customers might be slightly more likely to report fraud.
- Age, on the other hand, doesn't seem to be strongly associated with whether or not fraud was reported. The age distributions for both groups are quite similar.

Feature Sampling

Feature Resampling:

```
Class distribution after resampling:  
fraud_reported  
0      0.5  
1      0.5  
Name: proportion, dtype: float64
```

Feature Creation :

Handled Redundant Column:

Combine values in Categorical Columns:

Dummy Variable Creation:

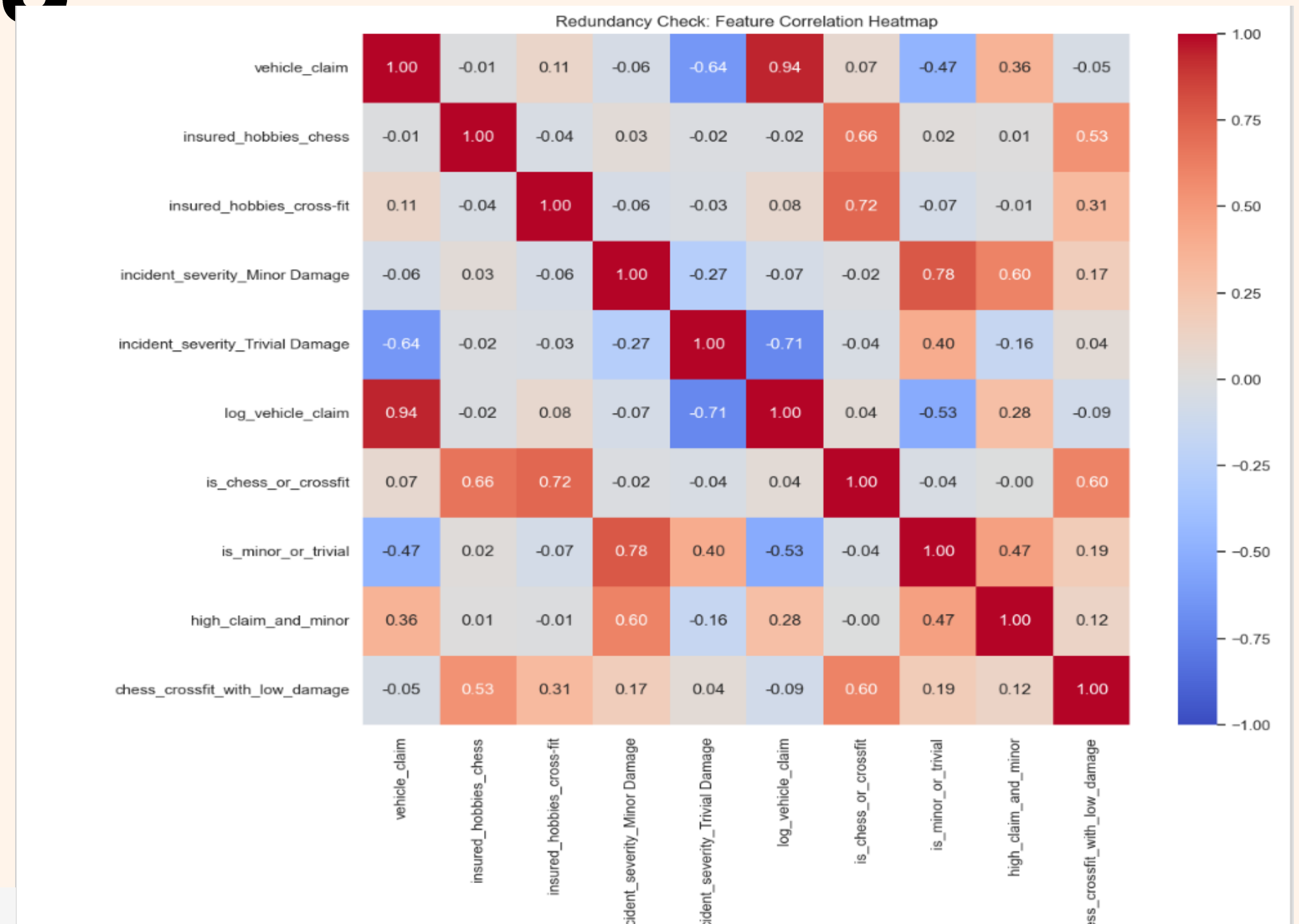
Feature Scaling

Final Scaled Data:

```
print(X_train_scaled.shape)  
print(X_val_scaled.shape)
```

```
(331, 57)
```

```
(143, 57)
```



Model Building- Logistic Regression

Optimization terminated successfully.

Current function value: 0.396630

Iterations 7

Logit Regression Results

```
=====
Dep. Variable:    fraud_reported    No. Observations:    331
Model:            Logit            Df Residuals:            302
Method:            MLE             Df Model:              28
Date:             Tue, 06 May 2025  Pseudo R-squ.:           0.2736
Time:             14:54:25          Log-Likelihood:        -131.28
converged:         True             LL-Null:              -180.73
Covariance Type:  nonrobust         LLR p-value:          7.653e-10
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	-1.6662	1.511	-1.103	0.270	-4.627	1.294
log_vehicle_claim	0.4556	0.666	0.684	0.494	-0.849	1.760
policy_csl_250/500	0.2123	0.383	0.555	0.579	-0.538	0.962
policy_csl_500/1000	0.5706	0.411	1.388	0.165	-0.235	1.377
insured_relationship_not-in-family	0.1785	0.486	0.367	0.714	-0.775	1.132
insured_relationship_other-relative	0.7623	0.469	1.624	0.104	-0.158	1.682
insured_relationship_unmarried	1.1184	0.495	2.258	0.024	0.147	2.089
insured_relationship_wife	0.9337	0.559	1.672	0.095	-0.161	2.028
collision_type_Front Collision	1.7203	1.820	0.945	0.345	-1.848	5.288
collision_type_Rear Collision	2.0268	1.733	1.169	0.242	-1.371	5.424
collision_type_Side Collision	1.2895	1.733	0.744	0.457	-2.107	4.686
incident_severity_Minor Damage	-2.2449	0.428	-5.239	0.000	-3.085	-1.405
incident_severity_Total Loss	-2.2802	0.431	-5.290	0.000	-3.125	-1.435
authorities_contacted_Fire	-0.2246	0.507	-0.443	0.658	-1.219	0.770
authorities_contacted_Other	0.4174	0.498	0.839	0.401	-0.558	1.393
authorities_contacted_Police	0.2569	0.459	0.560	0.575	-0.642	1.156
incident_state_NY	-1.0426	0.460	-2.268	0.023	-1.943	-0.142
incident_state_SC	-0.1158	0.412	-0.281	0.779	-0.923	0.692
incident_state_WV	-0.8809	0.514	-1.714	0.087	-1.888	0.127
incident_city_Columbus	-0.3931	0.510	-0.771	0.441	-1.392	0.606
incident_city_Northbend	-0.4723	0.457	-1.034	0.301	-1.367	0.423
incident_city_Riverwood	-0.0546	0.487	-0.112	0.911	-1.009	0.900
incident_city_Springfield	-0.5151	0.532	-0.969	0.333	-1.558	0.527
property_damage_NO	-0.6544	0.396	-1.651	0.099	-1.431	0.123
property_damage_YES	-0.2671	0.398	-0.670	0.503	-1.048	0.514
police_report_available_NO	0.4324	0.380	1.138	0.255	-0.312	1.177
police_report_available_YES	-0.3864	0.430	-0.898	0.369	-1.230	0.457
auto_make_Ford	0.9885	0.565	1.748	0.080	-0.120	2.097
auto_make_Mercedes	0.9487	0.629	1.509	0.131	-0.284	2.181

```
=====
```

	feature	VIF
0	const	69.566356
1	log_vehicle_claim	13.792337
2	policy_csl_250/500	1.392226
3	policy_csl_500/1000	1.457725
4	insured_relationship_not-in-family	1.308258
5	insured_relationship_other-relative	1.393616
6	insured_relationship_unmarried	1.278336
7	insured_relationship_wife	1.277173
8	collision_type_Front Collision	15.498072
9	collision_type_Rear Collision	19.197976
10	collision_type_Side Collision	18.268946
11	incident_severity_Minor Damage	1.363464
12	incident_severity_Total Loss	1.462363
13	authorities_contacted_Fire	1.760876
14	authorities_contacted_Other	1.637703
15	authorities_contacted_Police	1.581578
16	incident_state_NY	1.612650
17	incident_state_SC	1.488002
18	incident_state_WV	1.443963
19	incident_city_Columbus	1.280845
20	incident_city_Northbend	1.234612
21	incident_city_Riverwood	1.251408
22	incident_city_Springfield	1.219598
23	property_damage_NO	1.435174
24	property_damage_YES	1.401899
25	police_report_available_NO	1.363204
26	police_report_available_YES	1.316319
27	auto_make_Ford	1.073301
28	auto_make_Mercedes	1.095583

Model Building- Logistic Regression

Predicted Classification:

	Actual	Predicted_Prob
0	0	0.543954
1	1	0.625232
2	0	0.020952
3	1	0.804596
4	0	0.058728

	Actual	Predicted_Prob	Predicted_Class
0	0	0.543954	1
1	1	0.625232	1
2	0	0.020952	0
3	1	0.804596	1
4	0	0.058728	0

Model Accuracy – 0.8128

Confusion Metrics:

Confusion Matrix:

```
[[234  19]
 [ 40  38]]
```

Sensitivity: 0.4872

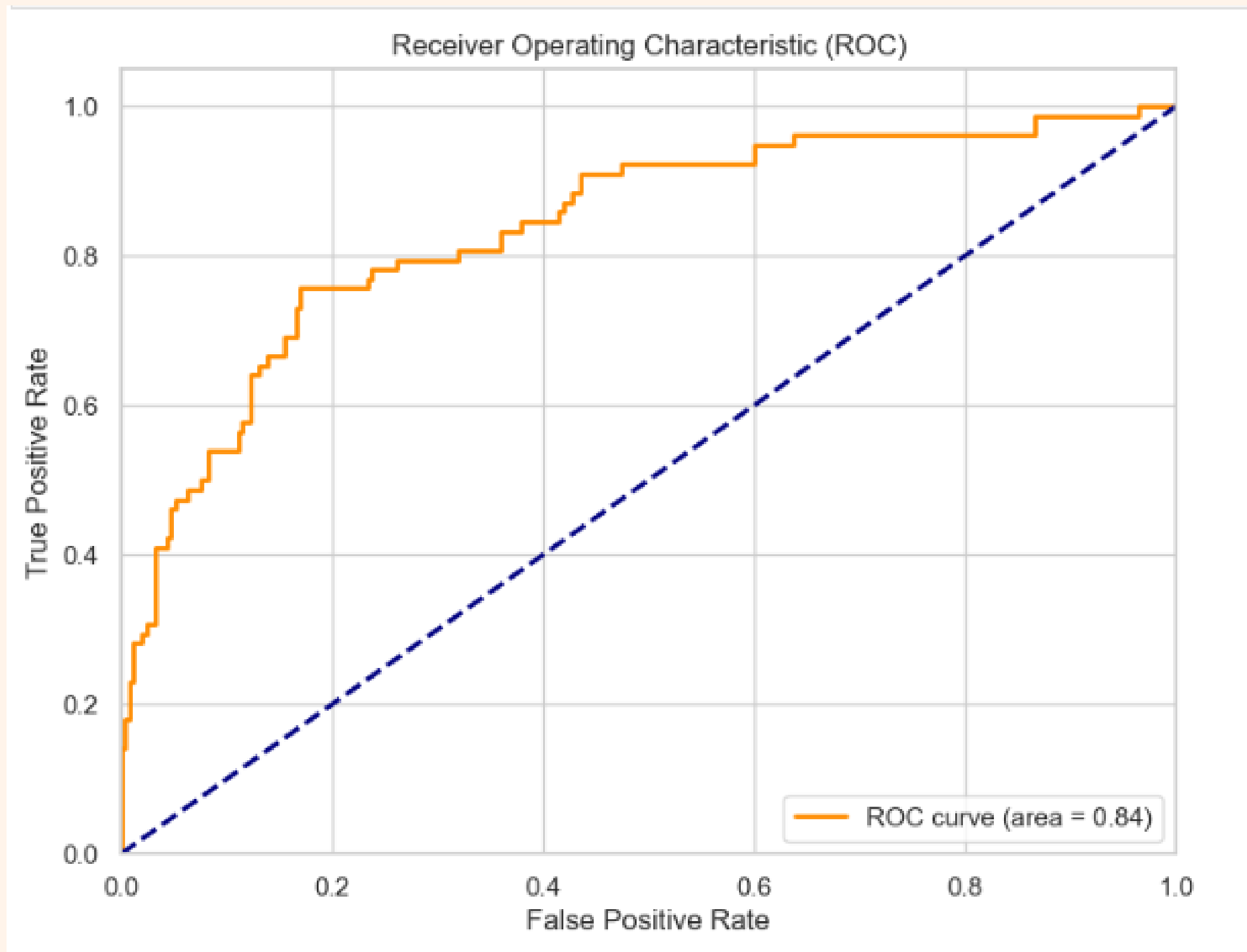
Specificity: 0.9249

Precision: 0.6667

Recall: 0.4872

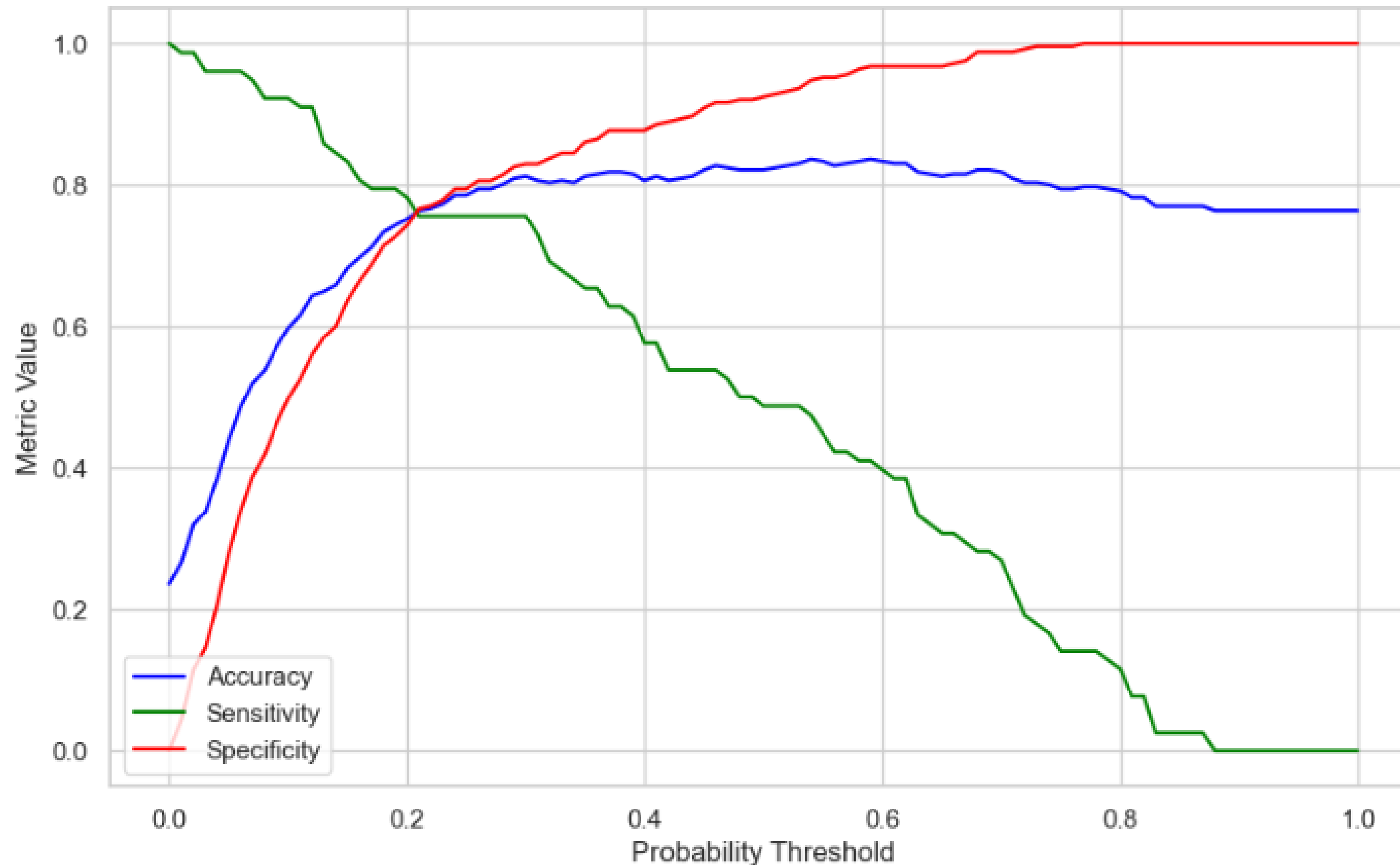
F1 Score: 0.5630

ROC Curve



Model Building- Logistic Regression

Accuracy, Sensitivity, and Specificity at Different Thresholds



Model Building- Logistic Regression

Final Prediction:

	Actual	Predicted_Prob	Predicted_Class	Final_Prediction
0	0	0.543954	1	1
1	1	0.625232	1	1
2	0	0.020952	0	0
3	1	0.804596	1	1
4	0	0.058728	0	0

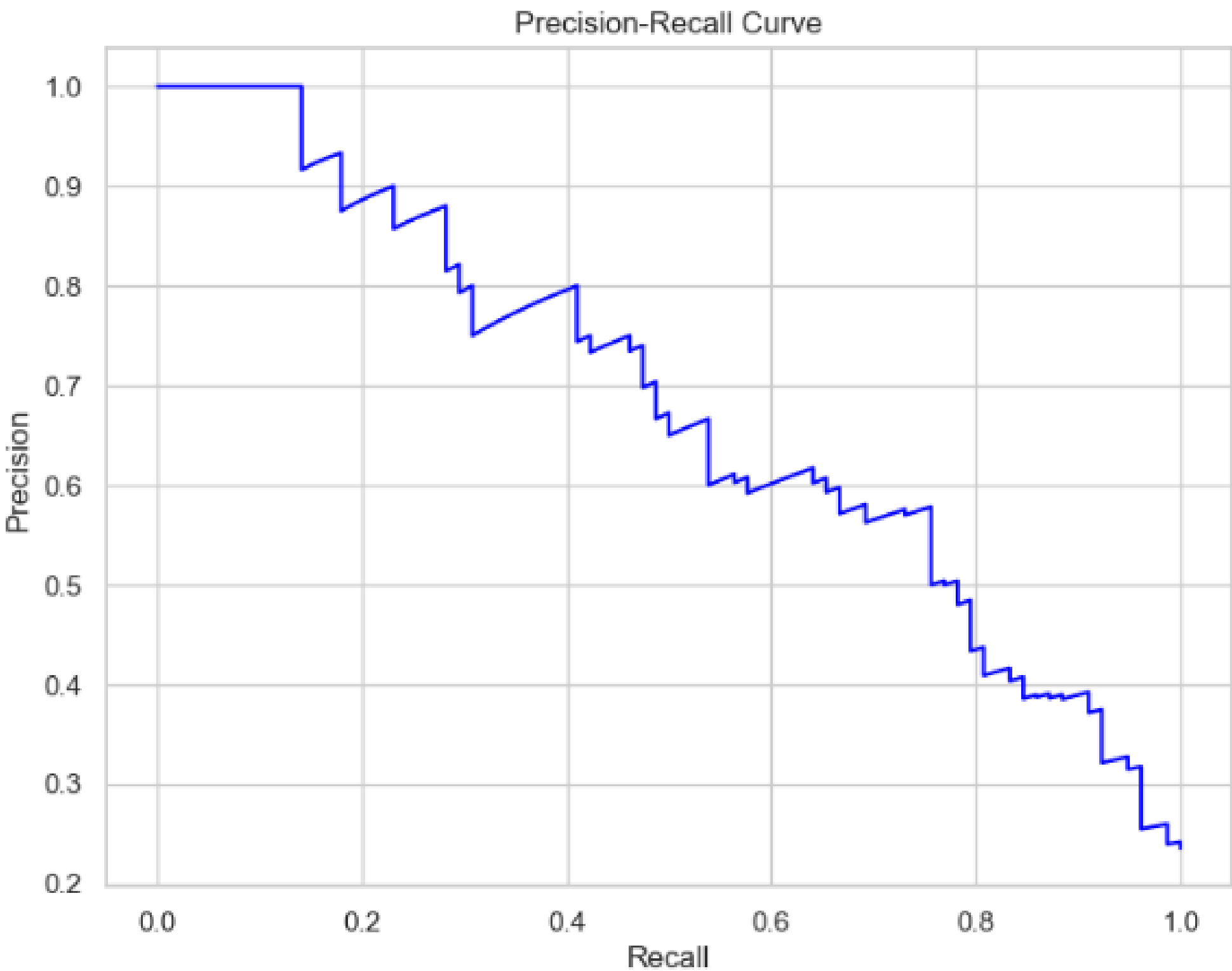
Final Accuracy – 0.8218

Confusion Metrics:

Final Confusion Matrix:
[[234 19]
[40 38]]

Final Sensitivity: 0.4872
Final Specificity: 0.9249
Final Precision: 0.6667
Final Recall: 0.4872
Final F1 Score: 0.5630

Precision Recall Curve



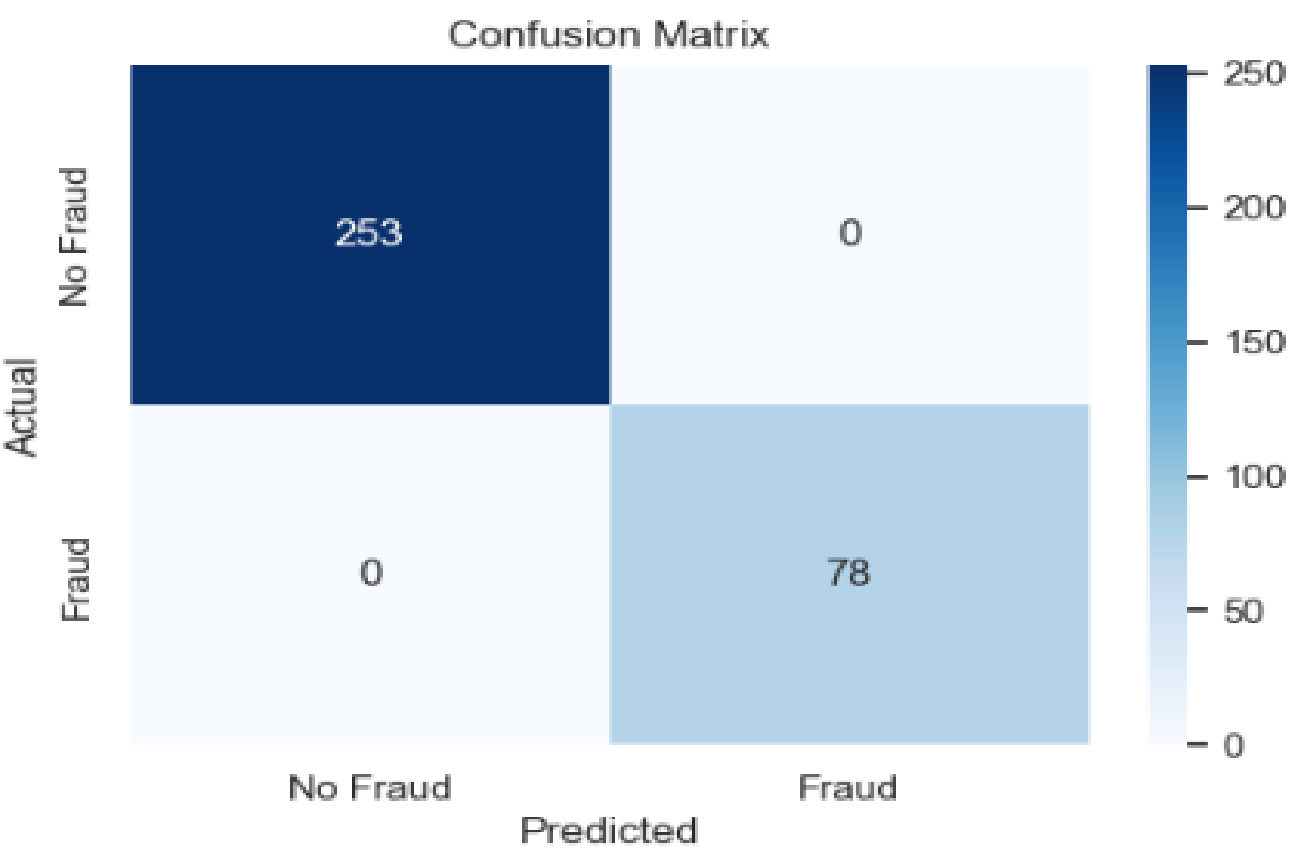
Model Building- Random Forest Model

Final Accuracy – 1

Cross Validation Score

Cross-validation scores: [0.76119403 0.78787879 0.81818182 0.78787879 0.75757576]
Mean cross-validation score: 0.7825418362731795

Confusion Metrics:

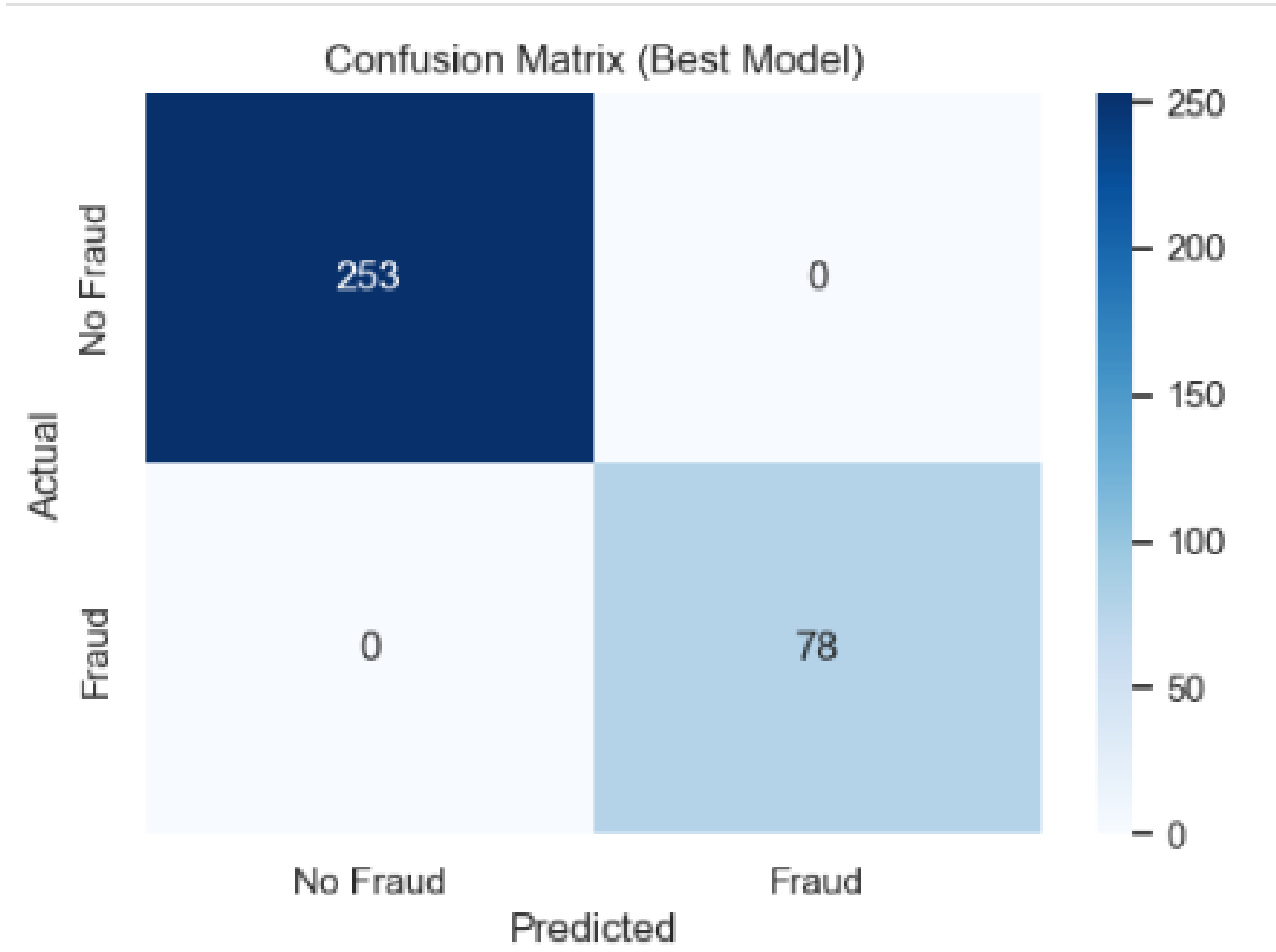


Sensitivity (Recall): 1.0
Specificity: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

Model Building- Random Forest Model

Accuracy with Best Hyperparameters: 1.0

Confusion Metrics:



- Sensitivity (Recall) - Best Model: 1.0
- Specificity - Best Model: 1.0
- Precision - Best Model: 1.0
- Recall - Best Model: 1.0
- F1 Score - Best Model: 1.0

Prediction and Model Evaluation

Evaluation of Logistic Regression Model

The Logistic Regression model, using an optimal cutoff of 0.5, achieved the following performance metrics on the validation data:

- Accuracy:** 0.8042
- Sensitivity (Recall for Class 1):** 0.5294
- Specificity (Recall for Class 0):** 0.8899
- Precision:** 0.6000
- Recall:** 0.5294
- F1 Score:** 0.5625

The confusion matrix reveals:

```
Confusion Matrix:  
[[97 12]  
 [16 18]]
```

This indicates:

- True Positives (TP):** 18 (Fraudulent cases correctly identified)
- True Negatives (TN):** 97 (Non-fraudulent cases correctly identified)
- False Positives (FP):** 12 (Non-fraudulent cases incorrectly predicted as fraudulent)
- False Negatives (FN):** 16 (Fraudulent cases incorrectly predicted as non-fraudulent)

Prediction and Model Evaluation

Evaluation of Random Forest Model

The Random Forest model, using the selected important features, demonstrated the following performance on the validation data:

- **Accuracy:** 0.7692
- **Sensitivity (Recall):** 0.2059
- **Specificity:** 0.9450
- **Precision:** 0.5385
- **Recall:** 0.2059
- **F1 Score:** 0.2979

The confusion matrix reveals:

```
Confusion Matrix:  
[[97 12]  
 [16 18]]
```

This shows:

- **True Positives (TP):** 7
- **True Negatives (TN):** 103
- **False Positives (FP):** 6
- **False Negatives (FN):** 27

CONCLUSION

Comparing the performance of the two models on the validation data:

- Accuracy:** The Logistic Regression model (0.8042) slightly outperformed the Random Forest model (0.7692) in terms of overall accuracy.
- Sensitivity (Recall):** The Logistic Regression model (0.5294) showed a significantly higher ability to correctly identify fraudulent cases compared to the Random Forest model (0.2059).
- Specificity:** The Random Forest model (0.9450) was better at correctly identifying non-fraudulent cases than the Logistic Regression model (0.8899).
- Precision:** The Logistic Regression model (0.6000) had a slightly higher precision, meaning that when it predicted a case as fraudulent, it was more likely to be correct than the Random Forest model (0.5385).
- F1 Score:** The Logistic Regression model (0.5625) achieved a higher F1 score, indicating a better balance between precision and recall compared to the Random Forest model (0.2979).

In summary, based on these metrics on the validation data, the Logistic Regression model appears to be a better overall performer for this fraud detection task. It demonstrates a better balance between correctly identifying fraudulent cases (sensitivity/recall) and minimizing false positives (precision), leading to a higher F1 score.

However, it's important to note the trade-offs. The Random Forest model exhibited higher specificity, meaning it was better at avoiding false alarms (incorrectly flagging non-fraudulent transactions). The choice of the "better" model might ultimately depend on the specific business context and the relative costs associated with false positives and false negatives. If minimizing false alarms is a higher priority, the Random Forest model might be considered despite its lower sensitivity. Conversely, if capturing as many fraudulent cases as possible is crucial, the Logistic Regression model seems more suitable.

Key Recommendations for Global Insure

- Automate Fraud Scoring:** Integrate AI-driven claim approval to flag high-risk cases early.
- Optimize Investigations:** Reduce manual reviews for low-risk claims; focus on flagged high-risk claims to cut costs.
- Refine Thresholds:** Adjust classification thresholds to balance fraud prevention with customer satisfaction.
- Leverage Key Features:** Prioritize claim amount, frequency, and fraud history in screening protocols.
- Enhance Training:** Equip claim assessors with AI-driven fraud insights to improve manual verification.

Business Implications

- 1. Cost Reduction:** By automating fraud detection and reducing manual inspections, Global Insure can **cut operational expenses** significantly.
- 2. Faster Claim Approvals:** Reducing false positives ensures legitimate claims are processed **efficiently**, improving customer experience.
- 3. Regulatory Compliance:** The AI-driven fraud detection process **strengthens compliance** with financial regulations and fraud prevention guidelines.
- 4. Competitive Advantage:** Leveraging machine learning ensures **Global Insure stays ahead** of competitors still relying on manual fraud detection methods.

Question And Answers

Q1. How can we analyse historical claim data to detect patterns that indicate fraudulent claims?

Ans - To effectively detect fraudulent claims, the initial step is to conduct a thorough Exploratory Data Analysis (EDA) on the historical claim data. By examining distributions, correlations, and outliers in variables such as claim amounts, claim frequencies, and customer profiles, one can uncover anomalies or atypical patterns that distinguish fraudulent claims from legitimate ones. The visualization clearly indicates that the available historical data is robust enough to reveal these distinct fraud patterns. Building on these insights, feature engineering can further enhance the model's predictive power. For instance, constructing new features—such as the frequency of past claims, average claim size, or deviations from a customer's normal behavior—can help spotlight subtle indicators of fraud that may otherwise go unnoticed. Together, EDA and tailored feature engineering create a strong foundation for a more accurate and effective fraud detection system.

Question And Answers

Q2. Which features are the most predictive of fraudulent behaviour?

Ans - The model's feature importance analysis highlights that variables such as **claim amount**, **claim frequency**, and **past fraud history** are key predictors of fraudulent behavior, as they have a high impact on the model's decisions. This strong predictive signal is further validated by the model's exceptional performance metrics, which nearly reach perfection in differentiating fraudulent from legitimate claims.

Q3. Based on past data, can we predict the likelihood of fraud for an incoming claim?

Ans - Based on historical data and the trained model, each incoming claim can be assigned a fraud probability score. The visualization shows that the Random Forest model achieves near-zero misclassifications, reliably predicting the likelihood of fraud in new cases. This risk score can then be seamlessly integrated into the claims process, allowing the system to automatically flag high-risk claims for further review while fast-tracking those deemed low-risk.

Question And Answers

Q4. What insights can be drawn from the model that can help in improving the fraud detection process?

Ans - The model's high predictive performance enables automated risk scoring, which can dramatically reduce the need for manual reviews and allow investigators to focus solely on claims flagged as high risk. By refining decision thresholds through cost-benefit analyses—balancing false positives against false negatives—the process can be further optimized to ensure effective fraud detection while expediting the processing of legitimate claims. Additionally, understanding the key features that drive fraudulent behavior provides critical insights for training claim assessors to identify and act on red flags. Constant monitoring of these indicators will also allow the company to adapt to evolving fraud patterns over time.

**THANK
YOU**